

Finite Scale Meshes Give the Exact Borel Complexity of Weak Octahedrality

Full solution packet for arXiv:2404.19457

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Status: full solution likely valid. In each of the Polish coding spaces $\mathcal{P}_\infty$ and $\mathcal{B}$, the isometry class $\widehat{\text{WOH}}$ of weakly octahedral separable Banach spaces is G_δ -complete. This improves the source paper's $F_{\sigma\delta}$ upper bound and answers its exact-complexity question in both coding spaces.

1 Source question

López-Pérez, Martínez Vañó, and Rueda Zoca prove that $\widehat{\text{WOH}}$ is $F_{\sigma\delta}$ in $\mathcal{P}_\infty$ and $\mathcal{B}$ [1, Theorem 6.6]. On source PDF pages 24–25 they ask for the exact complexity of weakly octahedral spaces. They explain that a positive answer to the related single-scale question would give a G_δ description in $\mathcal{B}$.

More precisely, the single-scale question asks whether weak octahedrality is equivalent to the following condition: for every $\varepsilon > 0$, every $x_1, \dots, x_n \in S_X$, and every $f \in B_{X^*}$, there is $y \in S_X$ such that

$$\|x_i + y\| \geq (1 - \varepsilon)(|f(x_i)| + 1) \quad (i = 1, \dots, n). \quad (1)$$

That characterization remains outside the scope of this packet. We instead settle the exact-complexity question directly.

It is then clear that $(\mu, g) \notin K$ because in that case g would be null for every $u \in N_\mu$, but we have that $\mu(e_1) = 0$ and $g(e_1) = 1$. This proves that K is not closed. ■

For these reasons, we were not able to directly study the isometry classes for the D2P and the DD2P in $\mathcal{P}_\infty$ and we left as an open question if this two classes are also G_δ in $\mathcal{P}_\infty$. In the case of the DD2P this is a very interesting question because a negative answer would prove that the DLD2P and the DD2P are different properties which, up to our knowledge, is an open question posed in [5, Question 4.1].

Lastly, as mentioned in Section 6, it is open if a weakly octahedral space X can be characterized as one satisfying that for any $\varepsilon > 0, n \in \mathbb{N}, x_1, \dots, x_n \in$

S_X and $f \in B_{X^*}$, there is a $y \in S_X$ such that for any $i \in \{1, \dots, n\}$ we have that

$$(7.7) \quad \|x_i + y\| \geq (1 - \varepsilon)(|f(x_i)| + 1).$$

Because of this, it is also an open question to get the exact complexity for weakly octahedral spaces. We were able to obtain a bound in Theorem 6.6 by using Szlenk derivatives, but this method does not seem to be able to produce a better result because of the comment previous to [11, Lemma 4.10] which asserts that the map s_2 used in the proof is almost optimal.

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Figure 1: The open questions in arXiv:2404.19457, source PDF pages 24–25.

2 Definitions and notation

All spaces are real. Recall the finite-set characterization [2, Proposition 2.2]: X is *weakly octahedral* if and only if, for every $\eta > 0, x_1, \dots, x_n \in S_X$, and $f \in B_{X^*}$, there is $y \in S_X$ such that

$$\|x_i + ty\| \geq (1 - \eta)(|f(x_i)| + t) \quad (i = 1, \dots, n, t > 0). \quad (2)$$

We use the source paper's coding. Let V be the countable rational vector space of finitely supported sequences and let μ be a seminorm code. The quotient completion is denoted by X_μ . The spaces $\mathcal{P}_\infty$ and $\mathcal{B}$ consist, respectively, of the codes for infinite-dimensional spaces and the codes for which μ is a norm on V ; see [3, 1]. For $v \in V$ with $\mu(v) > 0$, put

$$x_v^\mu = \frac{[v]}{\mu(v)} \in S_{X_\mu}.$$

The set of all such normalized rational directions is dense in S_{X_μ} .

3 Proof intuition

There are two compressions. First, at any requested accuracy, the continuum of scales $t > 0$ in (2) is unnecessary. The reverse triangle inequality handles very small and very large t . On the remaining compact interval, $t \mapsto \|x + ty\|$ is 1-Lipschitz, so a fixed finite rational mesh controls every scale.

Second, for finitely many test directions we do not need a global code for the entire dual ball. We retain only the finite evaluation vector $(f(x_1), \dots, f(x_n)) \in [-1, 1]^n$. The admissible evaluation vectors form a compact set with a closed graph: membership is exactly the countable family of Hahn–Banach inequalities for rational linear combinations. This remains true in $\mathcal{P}_\infty$, where the source paper’s global dual-ball graph is not closed. The finite-mesh witness relation is open; compactness of $[-1, 1]^n$ then makes its universal co-projection open. A countable intersection gives the desired G_δ definition.

4 The finite scale lemma

Fix an integer $m \geq 2$ and write $\varepsilon = 1/m$. Set

$$a_\varepsilon = \frac{\varepsilon}{4}, \quad b_\varepsilon = \frac{2}{\varepsilon}, \quad h_\varepsilon = \frac{\varepsilon^2}{64}.$$

Choose once and for all a finite rational h_ε -net T_ε in $[a_\varepsilon, b_\varepsilon]$.

Lemma 1 (finite scale mesh). *Let $x, y \in S_X$ and $c \in [0, 1]$. If*

$$\|x + sy\| \geq (1 - \varepsilon/2)(c + s) \quad (s \in T_\varepsilon), \quad (3)$$

then

$$\|x + ty\| \geq (1 - \varepsilon)(c + t) \quad (t > 0). \quad (4)$$

Proof. If $0 < t \leq a_\varepsilon$, then

$$\|x + ty\| \geq 1 - t \geq (1 - \varepsilon)(1 + t) \geq (1 - \varepsilon)(c + t).$$

The middle inequality follows by checking the right endpoint $t = \varepsilon/4$. If $t \geq b_\varepsilon$, then

$$\|x + ty\| \geq t - 1 \geq (1 - \varepsilon)(1 + t) \geq (1 - \varepsilon)(c + t),$$

because $\varepsilon(1 + t) \geq 2$.

It remains to take $t \in [a_\varepsilon, b_\varepsilon]$. Choose $s \in T_\varepsilon$ with $|t - s| \leq h_\varepsilon$. By (3) and the triangle inequality,

$$\begin{aligned} \|x + ty\| &\geq (1 - \varepsilon/2)(c + s) - h_\varepsilon \\ &\geq (1 - \varepsilon/2)(c + t) - 2h_\varepsilon. \end{aligned}$$

Since $c + t \geq a_\varepsilon = \varepsilon/4$ and $2h_\varepsilon = \varepsilon^2/32 \leq (\varepsilon/2)(c + t)$, this is at least $(1 - \varepsilon)(c + t)$. $\square$

Lemma 2 (joint continuity of finite-dimensional evaluation). *Fix a finite-dimensional coordinate subspace $E \subset c_{00}$. If $\mu_k \rightarrow \mu$ pointwise on V and $z_k \rightarrow z$ in E , then*

$$\bar{\mu}_k(z_k) \longrightarrow \bar{\mu}(z),$$

where $\bar{\mu}$ denotes the real extension of the rational seminorm. Consequently, all normalized norm expressions used below are continuous on the open set where their denominators are positive.

Proof. Choose coordinate vectors $e_1, \dots, e_d$ spanning E . Pointwise convergence makes $\sup_k \mu_k(e_j) < \infty$ for every j . Hence the $\bar{\mu}_k$ are equi-Lipschitz on E for the coordinate ℓ_1 norm. Approximate z by a rational vector $r \in E \cap V$, use pointwise convergence at r , and then use the uniform Lipschitz bound to compare z_k, r , and z . Letting the rational approximation error tend to zero proves the claim. $\square$

5 A robust countable criterion

Proposition 3. *A Banach space X is weakly octahedral if and only if the following holds. For every $m \geq 2$, every finite family $z_1, \dots, z_n$ from a fixed dense subset of S_X , and every $f \in B_{X^*}$, there is a y in that dense subset such that*

$$\|z_i + sy\| > (1 - \varepsilon/4)(|f(z_i)| + s) \quad (i = 1, \dots, n, s \in T_\varepsilon), \quad \varepsilon = 1/m. \quad (5)$$

Proof. Suppose first that X is weakly octahedral. Apply (2) to the z_i with error $\varepsilon/8$, obtaining $y_0 \in S_X$. Choose a dense point y with $\|y - y_0\| < \varepsilon/16$. For $s > 0$,

$$\begin{aligned} \|z_i + sy\| &\geq (1 - \varepsilon/8)(|f(z_i)| + s) - s\varepsilon/16 \\ &> (1 - \varepsilon/4)(|f(z_i)| + s), \end{aligned}$$

so (5) holds.

Conversely, let $x_1, \dots, x_n \in S_X$, $f \in B_{X^*}$, and a target error $\eta > 0$ be given. Choose m so large that $\varepsilon = 1/m < \eta$. Pick dense points z_i satisfying $\|x_i - z_i\| < \varepsilon^2/64$. Let y be supplied by (5). For $s \in T_\varepsilon$, using $|f(z_i) - f(x_i)| \leq \|z_i - x_i\|$, we get

$$\begin{aligned} \|x_i + sy\| &\geq \|z_i + sy\| - \|x_i - z_i\| \\ &> (1 - \varepsilon/4)(|f(x_i)| + s) - 2\|x_i - z_i\| \\ &\geq (1 - \varepsilon/2)(|f(x_i)| + s). \end{aligned}$$

The last step uses $s \geq a_\varepsilon = \varepsilon/4$. Lemma 1, with $c = |f(x_i)|$, now gives

$$\|x_i + ty\| \geq (1 - \varepsilon)(|f(x_i)| + t) > (1 - \eta)(|f(x_i)| + t)$$

for every i and $t > 0$. This is (2). □

6 Closed finite traces of the dual ball

Fix nonzero $v_1, \dots, v_n \in V$ and let

$$D_{\mathbf{v}} = \{\mu : \mu(v_i) > 0 \text{ for } i = 1, \dots, n\}.$$

This is open in both $\mathcal{P}_\infty$ and $\mathcal{B}$ (and equals all of $\mathcal{B}$ for a nonzero tuple). For $\mu \in D_{\mathbf{v}}$ define

$$C_{\mu, \mathbf{v}} = \{(f(x_{v_1}^\mu), \dots, f(x_{v_n}^\mu)) : f \in B_{X_\mu^*}\} \subset [-1, 1]^n.$$

Lemma 4. *The set*

$$\mathfrak{C}_{\mathbf{v}} = \{(\mu, \alpha) \in D_{\mathbf{v}} \times [-1, 1]^n : \alpha \in C_{\mu, \mathbf{v}}\}$$

is relatively closed. Equivalently, $\alpha \in C_{\mu, \mathbf{v}}$ if and only if, for every $q = (q_1, \dots, q_n) \in \mathbb{Q}^n$,

$$|\sum_{i=1}^n q_i \alpha_i| \leq \bar{\mu} \left(\sum_{i=1}^n \frac{q_i v_i}{\mu(v_i)} \right). \quad (6)$$

Proof. Every evaluation vector of a norm-one functional satisfies (6). Conversely, the inequalities show that the assignment $x_{v_i}^\mu \mapsto \alpha_i$ induces a well-defined linear functional of norm at most one on their span: norm-zero rational relations are respected, and real relations follow by density. Hahn–Banach extends it to $B_{X_\mu^*}$. Finally, Lemma 2 makes the right side of (6) continuous on $D_{\mathbf{v}}$. Thus the countable intersection of the closed inequalities (6) is relatively closed. □

7 The G_δ description

Fix m, n and $\mathbf{v} = (v_1, \dots, v_n)$. On $D_{\mathbf{v}} \times [-1, 1]^n$, let $R_{m,n,\mathbf{v}}$ consist of the pairs (μ, α) for which either $\alpha \notin C_{\mu,\mathbf{v}}$, or there is a $w \in V$ with $\mu(w) > 0$ such that, for every i and every $s \in T_\varepsilon$,

$$\bar{\mu}\left(\frac{v_i}{\mu(v_i)} + s \frac{w}{\mu(w)}\right) > (1 - \varepsilon/4)(|\alpha_i| + s). \quad (7)$$

By Lemmas 2 and 4, this relation is open: the inadmissibility alternative is open, and the witness alternative is a countable union over w of finite intersections of strict continuous inequalities.

Put

$$O_{m,n,\mathbf{v}} = \{\mu \in D_{\mathbf{v}} : \{\mu\} \times [-1, 1]^n \subset R_{m,n,\mathbf{v}}\}.$$

Since the cube is compact and $R_{m,n,\mathbf{v}}$ is open, $O_{m,n,\mathbf{v}}$ is open relative to $D_{\mathbf{v}}$ (the tube lemma). Therefore

$$S_{m,n,\mathbf{v}} = D_{\mathbf{v}}^c \cup O_{m,n,\mathbf{v}}$$

is G_δ : in a metric space the union of a closed set and an open set is G_δ .

Theorem 5. *In both $\mathcal{P}_\infty$ and $\mathcal{B}$,*

$$\widehat{\text{WOH}} = \bigcap_{m \geq 2} \bigcap_{n \geq 1} \bigcap_{\mathbf{v} \in (V \setminus \{0\})^n} S_{m,n,\mathbf{v}}. \quad (8)$$

Consequently, $\widehat{\text{WOH}}$ is a G_δ set in each coding space.

Proof. For a tuple whose entries survive the quotient, Lemma 4 runs through exactly all possible restrictions of $f \in B_{X_\mu^*}$ to the corresponding normalized directions. Condition $\mu \in O_{m,n,\mathbf{v}}$ therefore says exactly that the finite-mesh criterion (5) holds for that rational-direction tuple. Tuples with a zero quotient direction impose no condition, as recorded by $D_{\mathbf{v}}^c$. The normalized surviving rational directions are dense in S_{X_μ} . Proposition 3 now proves (8). All three index sets on the right are countable, and each $S_{m,n,\mathbf{v}}$ is G_δ , so their intersection is G_δ . $\square$

Theorem 6 (exact complexity). *The class $\widehat{\text{WOH}}$ is G_δ -complete in $\mathcal{P}_\infty$ and in $\mathcal{B}$.*

Proof. The source paper proves that $\widehat{\text{WOH}}$ is not F_σ in either coding space [1, Proposition 3.3]. Theorem 5 gives the G_δ upper bound. The standard Hurewicz completeness theorem for non- F_σ G_δ subsets of Polish spaces [4, Theorem 22.11] therefore makes $\widehat{\text{WOH}}$ G_δ -complete in each space. $\square$

8 Verification and limitations

The proof has no computational dependency. Its quantitative interface is Lemma 1; all constants have slack by factors of at least two. The descriptive-set-theoretic interface is finite-dimensional: the compact cube $[-1, 1]^n$ replaces the nonclosed global dual-ball graph in $\mathcal{P}_\infty$. The Hahn–Banach inequalities (6) verify that no dual restrictions are lost or added.

A bounded novelty search on 22 July 2026 covered the run’s four cheap indexes; the exact arXiv id and title; searches for weak octahedrality with “finite mesh”, “ G_δ ”, and “Borel complexity”; arXiv records and the current source version; and the cited weak-octahedrality characterization. No later paper or preprint stating this exact-complexity result was found. Novelty confidence is moderate: the argument is short and may be independently accessible, but the source explicitly leaves the complexity open and its stated obstruction is bypassed here.

This packet settles the exact Borel complexity in both coding spaces named by the source theorem. It does *not* prove or disprove the single-scale characterization (1). Human review should focus on the relative closedness of $\mathfrak{C}_{\mathbf{v}}$ when quotient kernels vary, and on the compact co-projection step defining $O_{m,n,\mathbf{v}}$.

References

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